UniStory: A Unified Autoregressive–Diffusion Transformer for Visual-Textual Story Generation

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1 Problem and Design Overview

Given a short prompt c (e.g., *"We build a Disney park on the moon"*), we model the joint distribution $p_\theta(s, I \mid c)$, where $s = (s_1, \ldots, s_N)$ is a long narrative and $I \in \mathbb{R}^{H \times W \times 3}$ is a representative image. Text is discrete and sequential; images are continuous and spatial—a single model must accommodate both. Moreover, c typically underspecifies visual content, so the image should reflect the *generated story*, not only the prompt. We propose **UniStory**, a single Transformer that performs next-token prediction on text and denoising diffusion on continuous image latents within a shared sequence, following the Transfusion paradigm [Zhou et al., 2024]. A two-stage *sketch-then-expand* decoder improves long-story coherence, and a CLIP-based alignment loss [Radford et al., 2021] ties the image to the story.

2 Model Architecture

Figure 1 illustrates the pipeline. A BPE tokenizer embeds the prompt and the story, while a pre-trained and *frozen* VAE encoder E maps an image to latent patches $z_0 = E(I) \in \mathbb{R}^{h \times w \times d}$. Lightweight modality adapters project both into a shared d-dimensional space; the same Transformer then consumes a *mixed sequence* of text tokens and image patches. The attention mask is causal on text but bidirectional within the image block, preserving text→image causal order globally [Zhou et al., 2024]. Two output heads branch from the Transformer: a softmax *text head* over the vocabulary (AR), and a *diffusion head* that predicts ϵ. A frozen VAE decoder D maps the denoised latent to pixels; only the Transformer, adapters, and heads are trainable. For long-story coherence, text decoding proceeds as *two sequential AR passes sharing the same weights and a single forward graph*: using learned separator tokens $\langle\text{SK}\rangle, \langle\text{FS}\rangle$, the model consumes $[c; \langle\text{SK}\rangle; \tilde{s}; \langle\text{FS}\rangle; s]$ and emits a sketch $\tilde{s}$ (≤ 50 tokens) after $\langle\text{SK}\rangle$ and the full story s (~ 500 tokens) after $\langle\text{FS}\rangle$.

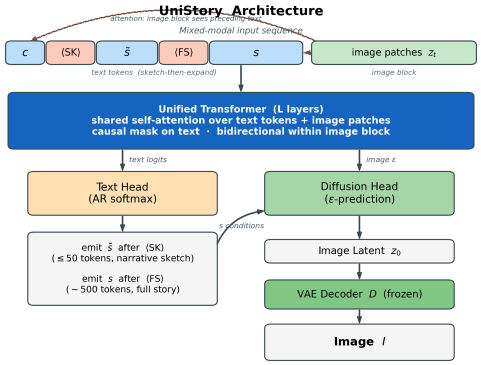

Figure 1: UniStory interleaves text tokens and image patches in one Transformer. Learned separators $\langle\text{SK}\rangle, \langle\text{FS}\rangle$ split the text side into sketch-then-expand passes; the image block attends to all preceding text, so the diffusion head is conditioned on c and the generated story s.

3 Training Objectives

Let $x = (c, s, z_0)$ be a training triple. The **autoregressive text loss** (chain-rule factorization of the joint probability) is

$$\mathcal{L}_{\text{LM}}(\theta) \;=\; -\mathbb{E}_{(c,s)}\left[\sum_{t=1}^{N} \log p_\theta\big(s_t \mid s_{<t},\, c\big)\right]. \tag{1}$$

The **diffusion loss** on image latents follows DDPM [Ho et al., 2020], an ELBO on the log-likelihood. With forward process $q(z_t \mid z_0) = \mathcal{N}(z_t; \sqrt{\bar{\alpha}_t}\, z_0, (1-\bar{\alpha}_t)\mathbf{I})$, the network ϵ_θ predicts the injected noise, conditioned on both the prompt c and the generated story s:

$$\mathcal{L}_{\text{diff}}(\theta) \;=\; \mathbb{E}_{z_0,\,\epsilon,\,t}\Big[\,\big\|\,\epsilon - \epsilon_\theta(z_t, t, c, s)\,\big\|_2^2\Big], \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}),\; t \sim \mathcal{U}\{1, \ldots, T\}. \tag{2}$$

A **cross-modal alignment loss** pulls the generated image toward the story in CLIP space with frozen encoders $\phi_{\text{txt}}, \phi_{\text{img}}$. To keep it differentiable and cheap, we avoid running the full reverse chain and instead use the x_0-*prediction shortcut*: at the same sampled t as in $\mathcal{L}_{\text{diff}}$, we reconstruct the clean latent in one step from ϵ_θ:

$$\hat{z}_0 \;=\; \frac{z_t - \sqrt{1-\bar{\alpha}_t}\;\epsilon_\theta(z_t, t, c, s)}{\sqrt{\bar{\alpha}_t}}, \qquad \mathcal{L}_{\text{align}}(\theta) \;=\; 1 \;-\; \cos\!\big(\phi_{\text{txt}}(s),\; \phi_{\text{img}}(D(\hat{z}_0))\big). \tag{3}$$

Total objective:

$$\boxed{\mathcal{L}_{\text{UniStory}}(\theta) \;=\; \mathcal{L}_{\text{LM}} \;+\; \lambda_{\text{diff}}\,\mathcal{L}_{\text{diff}} \;+\; \lambda_{\text{align}}\,\mathcal{L}_{\text{align}}} \quad (\lambda_{\text{diff}}{=}1.0,\; \lambda_{\text{align}}{=}0.1). \tag{4}$$

4 Training Strategy and Inference

Training proceeds in two stages. *Stage 1 (pretraining)*: large-scale image–caption pairs (e.g., LAION-5B) and text-only corpora are used jointly with $\mathcal{L}_{\text{LM}}$ and $\mathcal{L}_{\text{diff}}$; single-modality batches activate only the corresponding term. *Stage 2 (fine-tuning)*: prompt–story pairs (c, s) are harvested from the Reddit WritingPrompts corpus (the top-level comment acts as s, the title as c); for each pair we synthesize an illustration I by running a frozen pretrained text-to-image teacher (e.g., SDXL) on s, and retain triples with $\text{CLIPScore}(s, I) \geq 0.3$ (approximately the top 60%); $\mathcal{L}_{\text{align}}$ is activated only in this stage. We use AdamW with cosine schedule, classifier-free guidance (CFG) dropout of c and s at 10% each, and QK-Norm with an auxiliary z-loss to stabilize mixed-modal training [Chameleon Team, 2024].

Inference: (i) sample sketch $\tilde{s} \sim p_\theta(\cdot \mid c)$; (ii) sample full story $s \sim p_\theta(\cdot \mid c, \tilde{s})$ with nucleus sampling (p=0.9); (iii) from $z_T \sim \mathcal{N}(0, \mathbf{I})$, run T reverse diffusion steps with CFG, $\hat{\epsilon} = (1{+}w)\,\epsilon_\theta(z_t, t, c, s) - w\,\epsilon_\theta(z_t, t, \varnothing)$ where $\varnothing$ denotes dropping both c and s and w=3.0; decode $I = D(z_0)$.

Design justification. Pure AR image models (e.g., Chameleon) rely on VQ tokenization that caps fidelity; pure diffusion cannot natively generate long coherent text. The hybrid design keeps each paradigm's strength [Zhang et al., 2025]; conditioning the image on s rather than c captures concrete visual elements (e.g., *"Space Princess Castle"*) that only emerge during story generation.

References

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